Rotary Wing Aerodynamics (AE 667)



Vertical Descent of Rotors

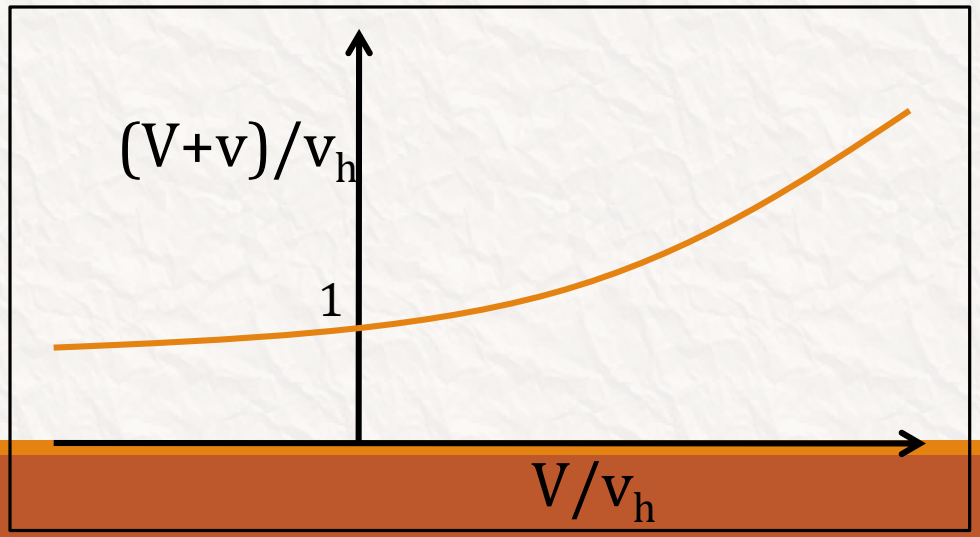
Using Momentum Theory

- Expression for power was:

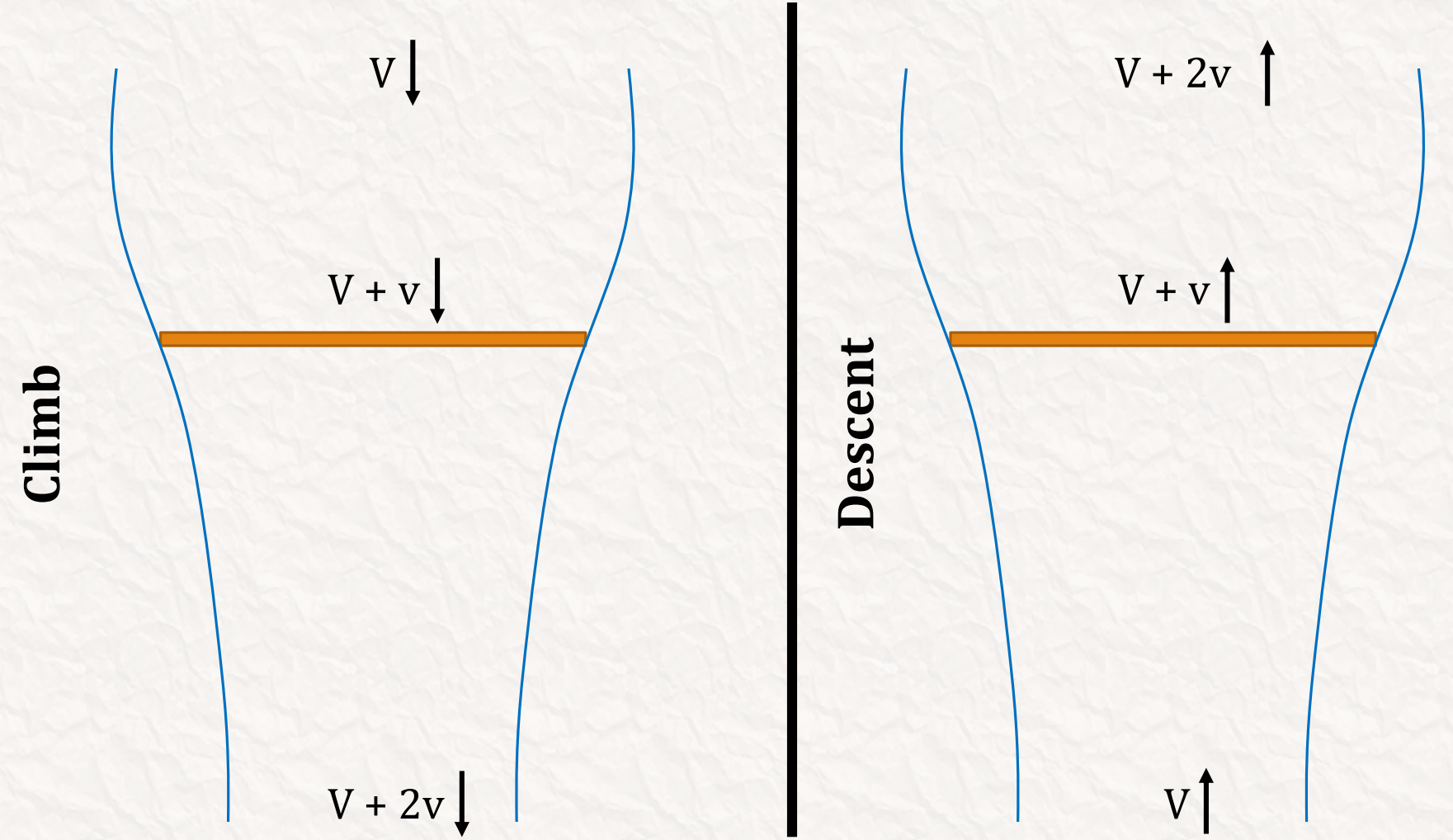
$$P = T\left[\frac{V}{2} + \sqrt{\left(\frac{V}{2}\right)^2 + \frac{T}{2\rho A}}\right]$$

- P never goes to 0 or negative, irrespective of descent rate

So what??



Using Momentum Theory



Using Momentum Theory

- Recall:

$$\mathrm{v} = -\frac{\mathrm{V}}{2} + \sqrt{\left(\frac{V}{2}\right)^2 + \frac{T}{2\rho A}}$$ (for positive climb)

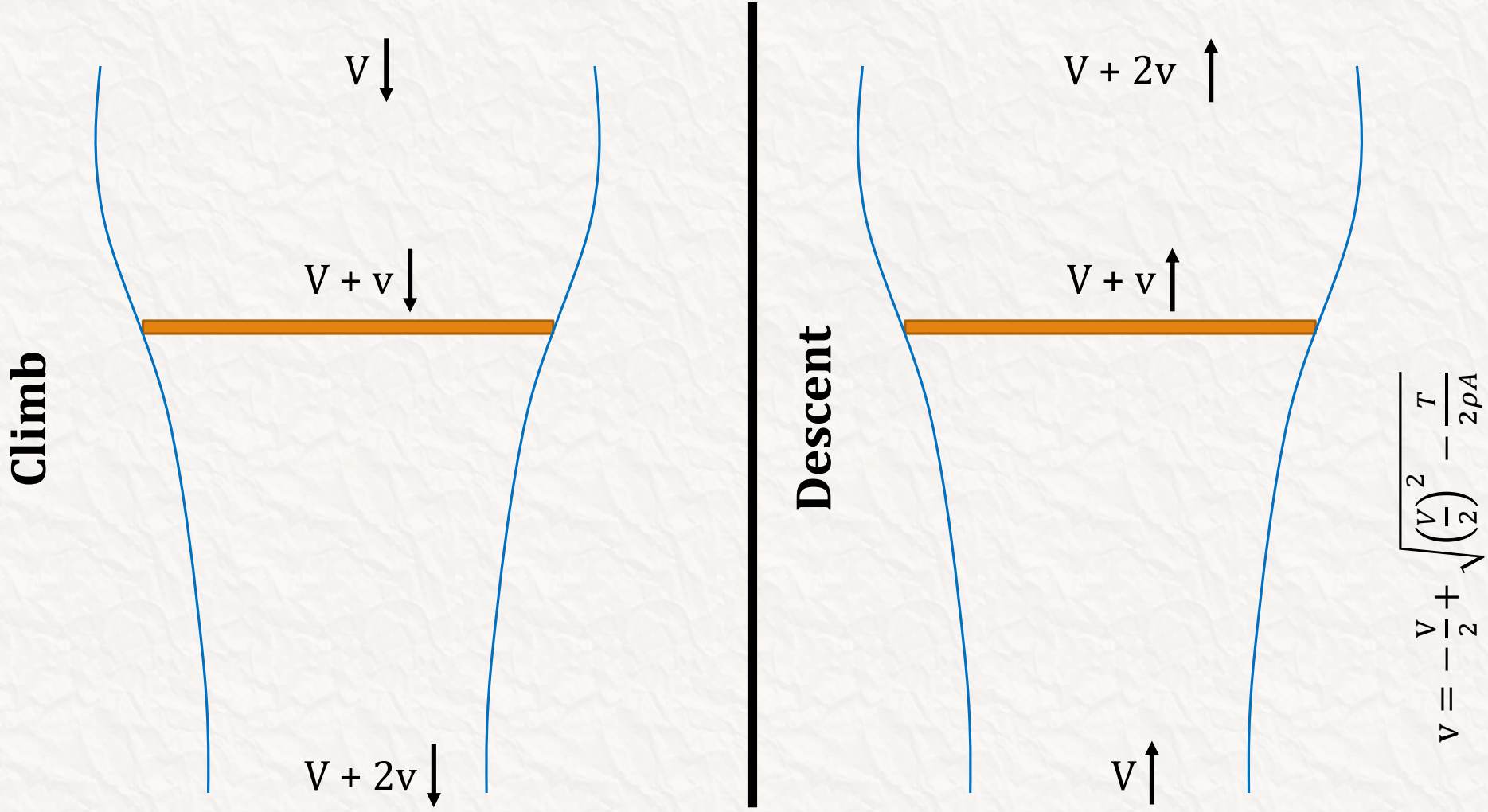
- But here in descent:

$$\mathrm{v} = -\frac{\mathrm{V}}{2} - \sqrt{\left(\frac{V}{2}\right)^2 - \frac{T}{2\rho A}}$$ (for descent, hence V < 0)

- $$\mathrm{v} = -\frac{\mathrm{V}}{2} + \sqrt{\left(\frac{V}{2}\right)^2 - \frac{T}{2\rho A}}$$ is discarded..

Why?

Using Momentum Theory



$$v = -\frac{V}{2} + \sqrt{\left(\frac{V}{2}\right)^2 - \frac{T}{2\rho A}}$$

Rewriting the same differently:

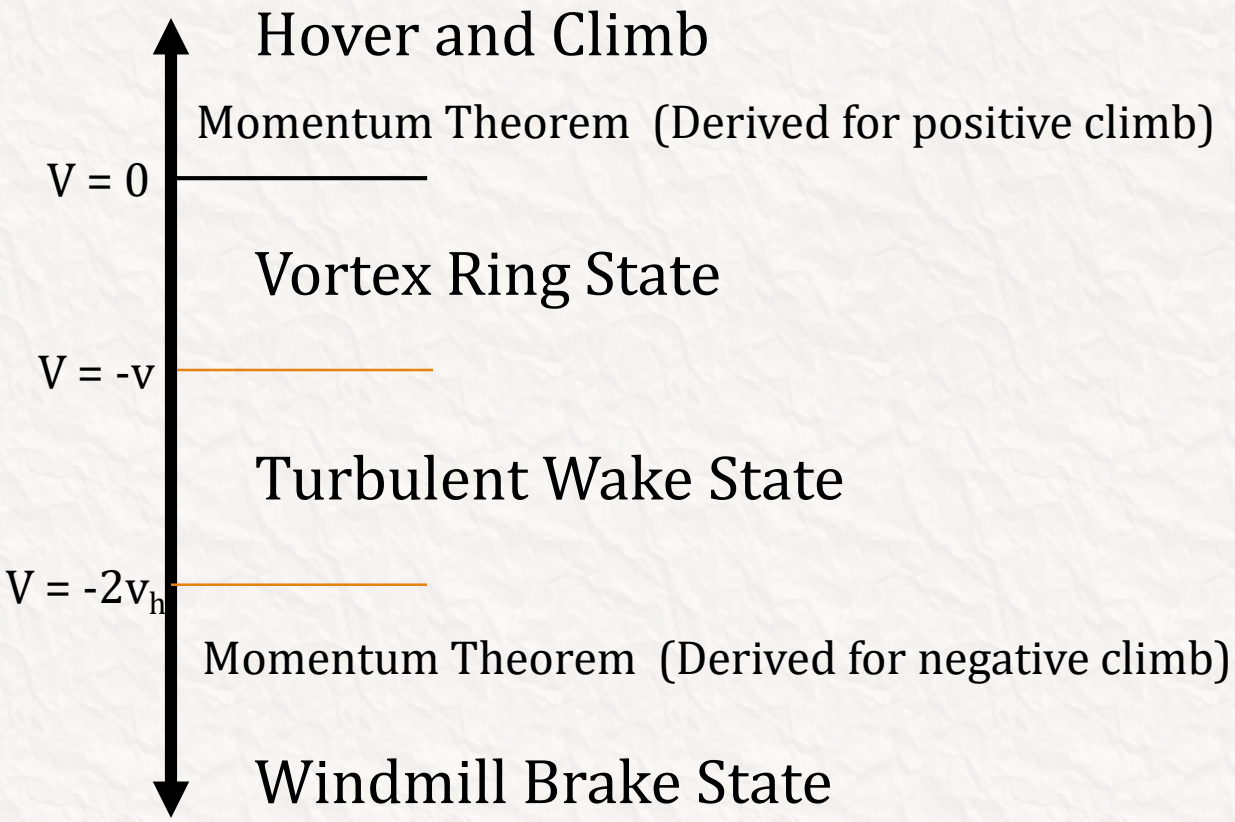
- $\frac{v}{v_h} = -\frac{V}{2v_h} + \sqrt{\left(\frac{V}{2v_h}\right)^2 + 1}$ for climb ($V \geq 0$)
- $\frac{v}{v_h} = -\frac{V}{2v_h} - \sqrt{\left(\frac{V}{2v_h}\right)^2 - 1}$ for descent ($V < 0$)

  [Good only for $V < -2v_h$]

$$\frac{v}{v_h} = -\frac{V}{2v_h} + \sqrt{\left(\frac{V}{2v_h}\right)^2 - 1}$$

[Not possible as it will result in $v/v_h > 1$, which will cause flow above the rotor (V+2v) to be in the downward direction]

States of Vertical Descent



States of Vertical Descent

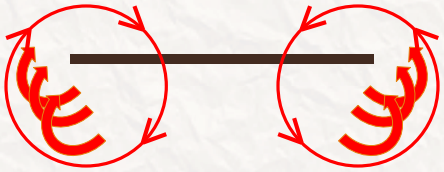
Momentum Theory not valid for:

- Vortex Ring State
- Turbulent Wake State

Because definite slip stream ceases to exist

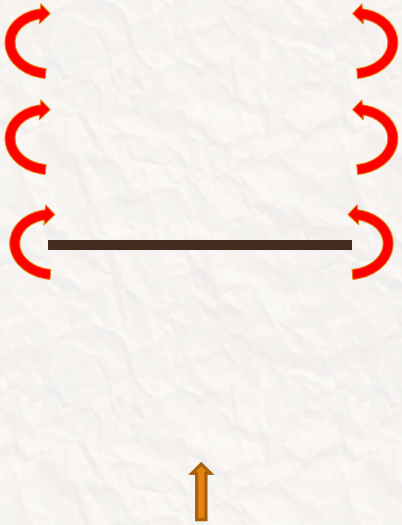
Because the flow inside the slip stream changes direction

States of Vertical Descent



Vortex Ring State

V↑ V+v↓ V+2v↓



Turbulent Wake State

V↑ V+v↑ V+2v↓

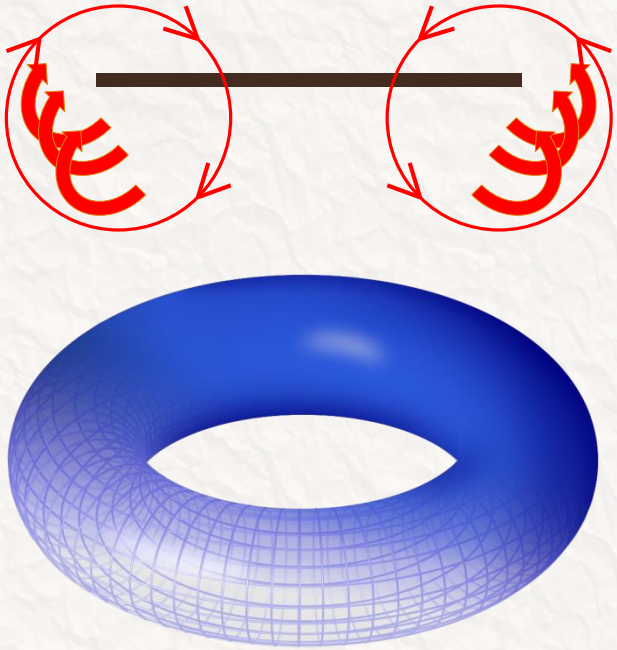


Wind Mill Brake State

V↑ V+v↑ V+2v↑

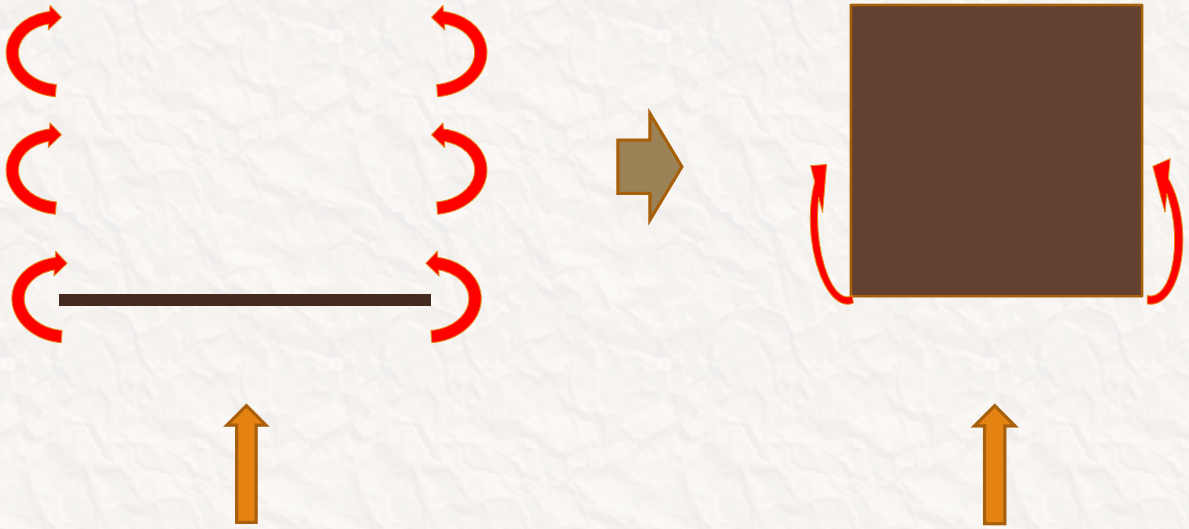
Vortex Ring State

- Vortices pile up on each other and form a torus shaped ring due to air from bottom
- Ring grows and bursts periodically
- Causes loss of thrust on rotor (sometimes leading to accidents)
- Unsteady flow
- Analytical analysis difficult. Rely on empirical predictions



Turbulent Wake State

- Rotor behaves like a bluff body
- Flow unsteady, rely on empirical means



Vortex Ring State



Wind Mill Brake State

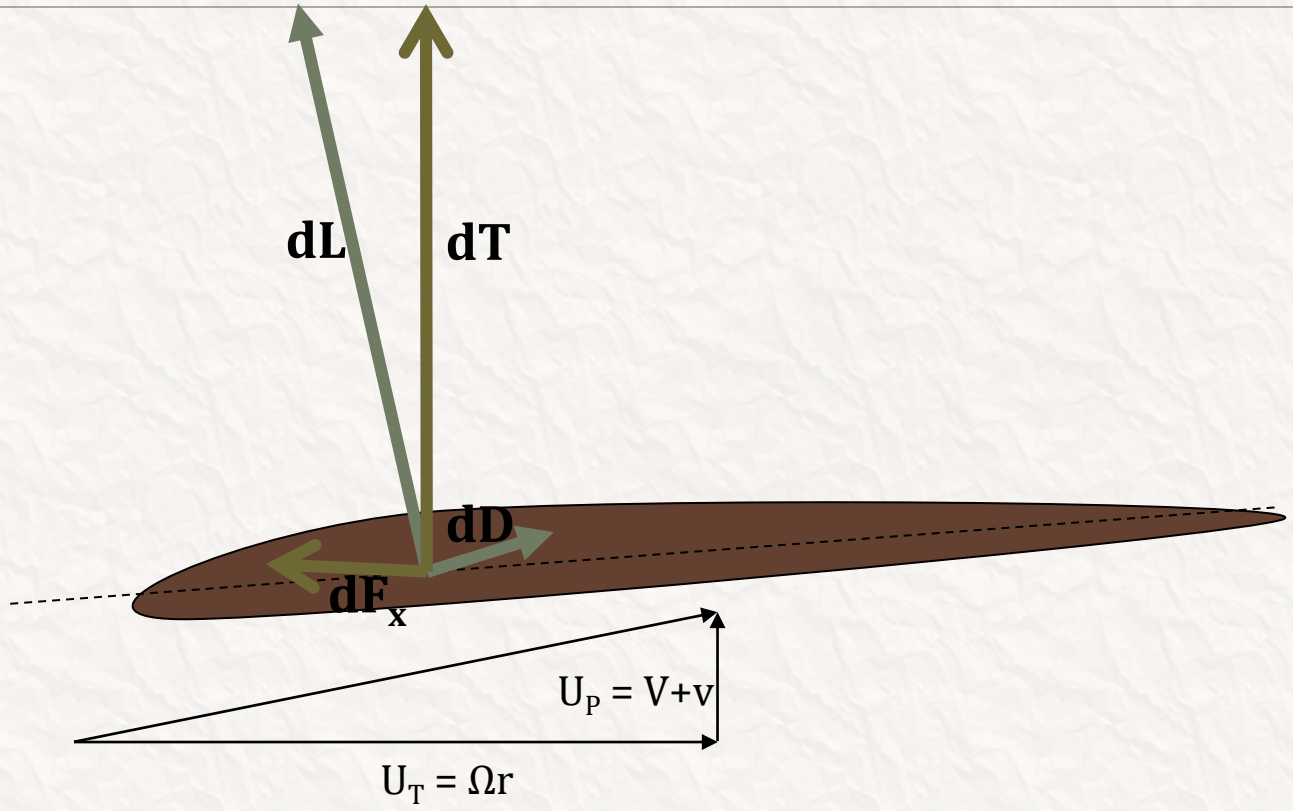
- Power extracted from the flow
- Steady, hence momentum theory can be used
- T = - 2ρAv(V+v)
  [notice the '-' sign]



$$\therefore \mathrm{v} = -\frac{\mathrm{V}}{2} - \sqrt{\left(\frac{V}{2}\right)^2 - \frac{T}{2\rho A}}; \quad P = T(V + \mathrm{v})$$

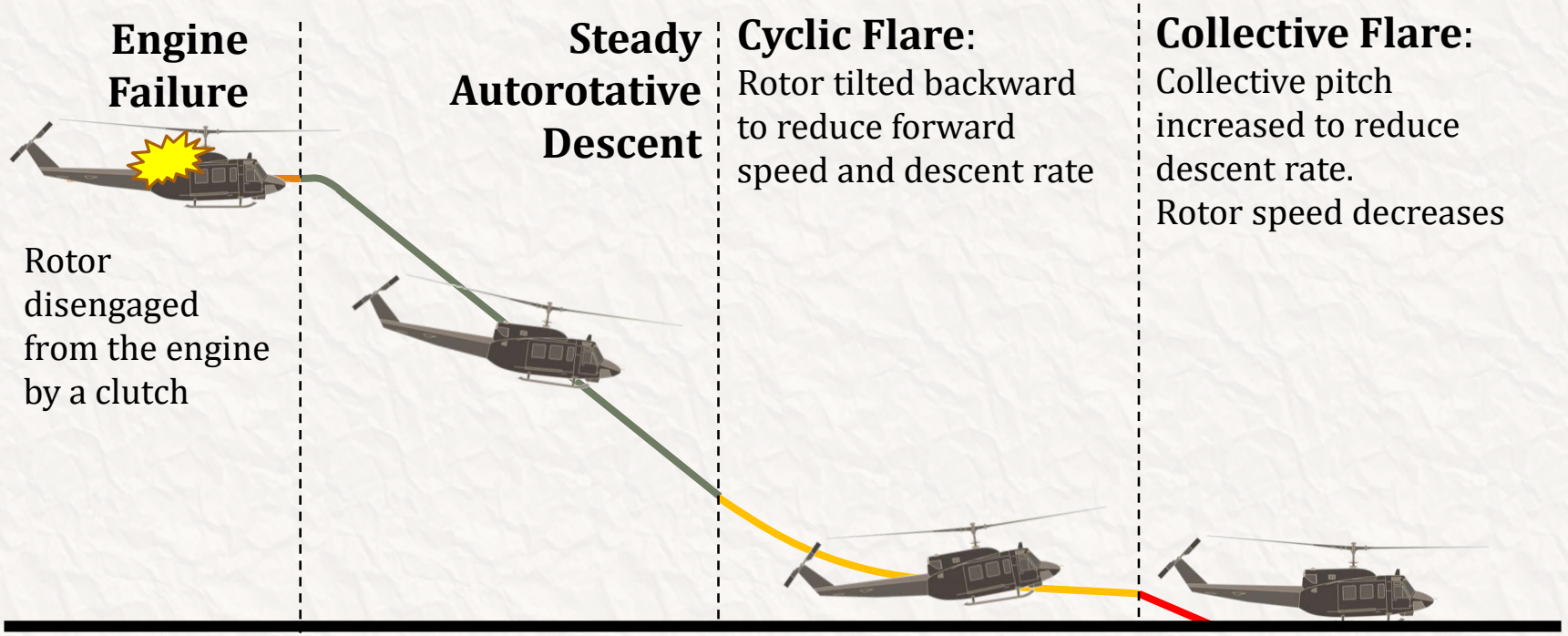
In this case, $P < 0$ means power is extracted, from the flow, like a wind mill.

Wind Mill Brake State



Forward directed dF_x (induced thrust?) rotates the blade against drag. Blade still generates some thrust dT

Helicopter Autorotative Landing



Helicopter Autorotative Landing



For Steady Autorotative Vertical Descent

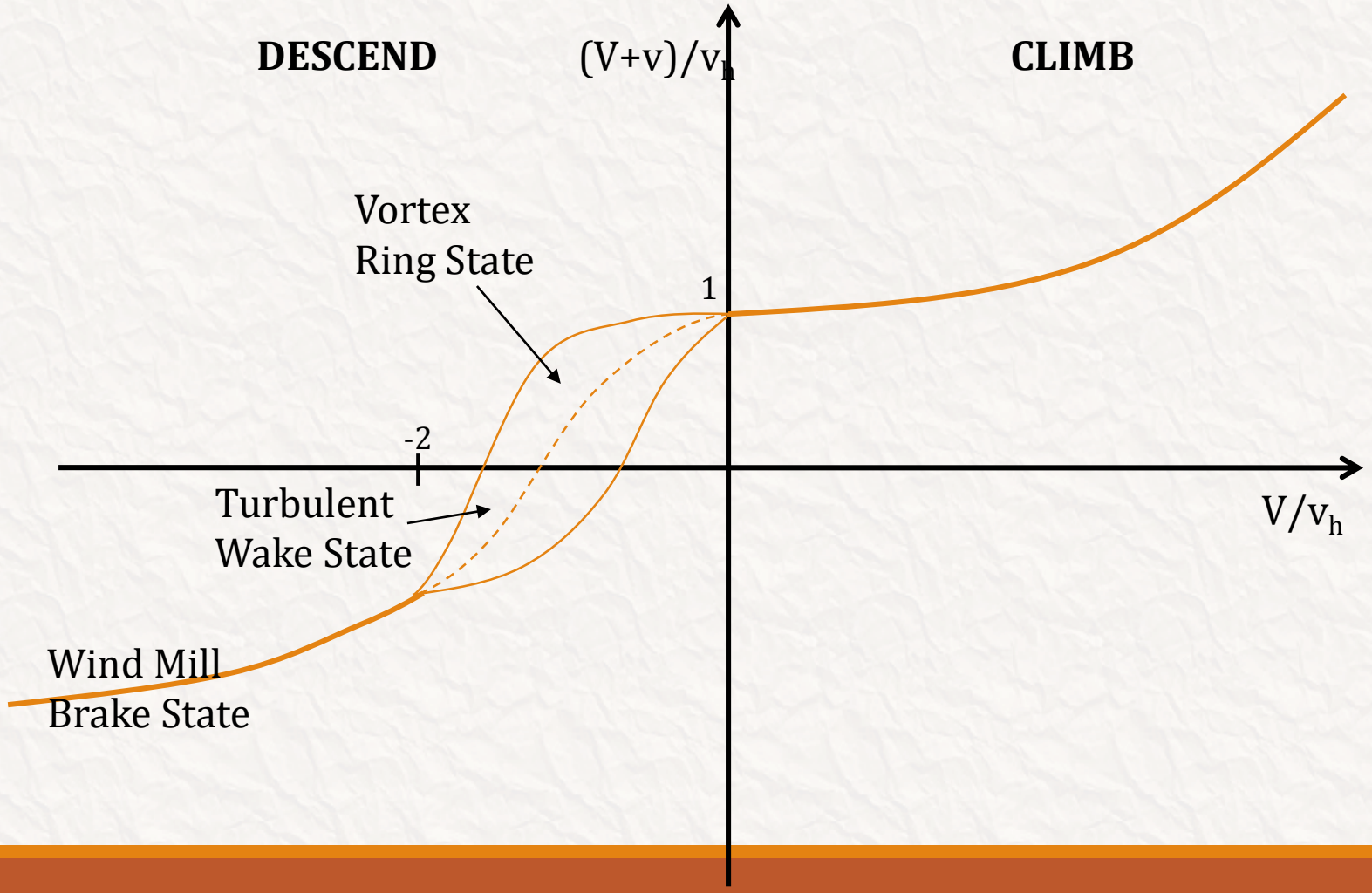
- Shaft power = climb power + induced power + profile power

$$P = TV + Tv + \frac{\sigma C_{d0}}{8}\rho\pi R^2(\Omega R)^3$$

- But the engine is off, so….
- P = 0

$$TV = -Tv - \frac{\sigma C_{d0}}{8}\rho\pi R^2(\Omega R)^3$$

All states in a plot



All states in a plot

